

UNIVERSITÀ
di **VERONA**

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HCI – intelligent *multimodal* interfaces

Feature matching

Master program in Artificial Intelligence

Convolution

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau \text{ for } f, g : [0, \infty] \rightarrow \mathbb{R}$$

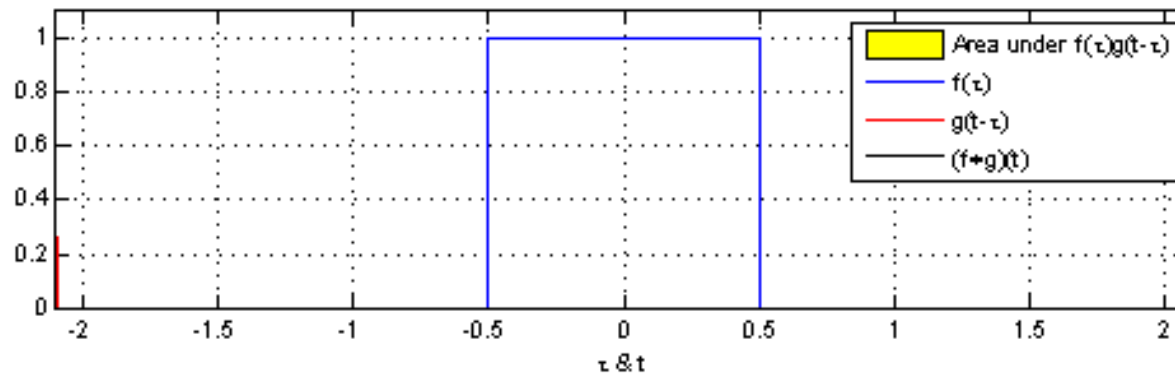
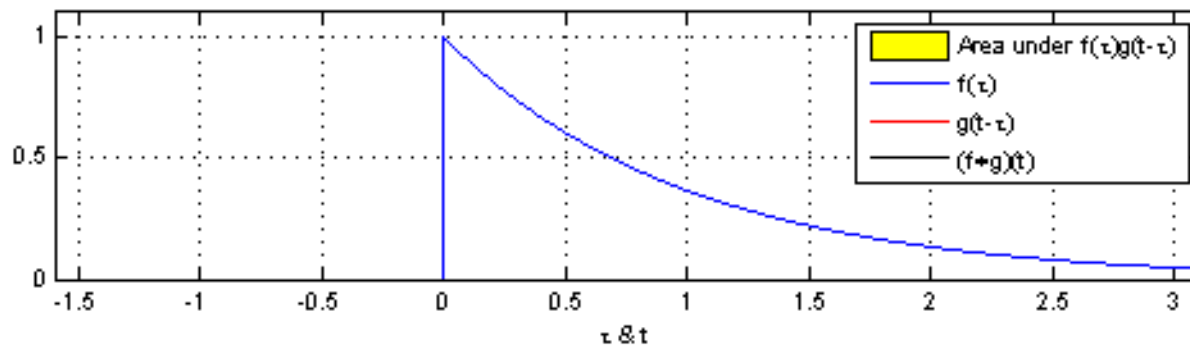
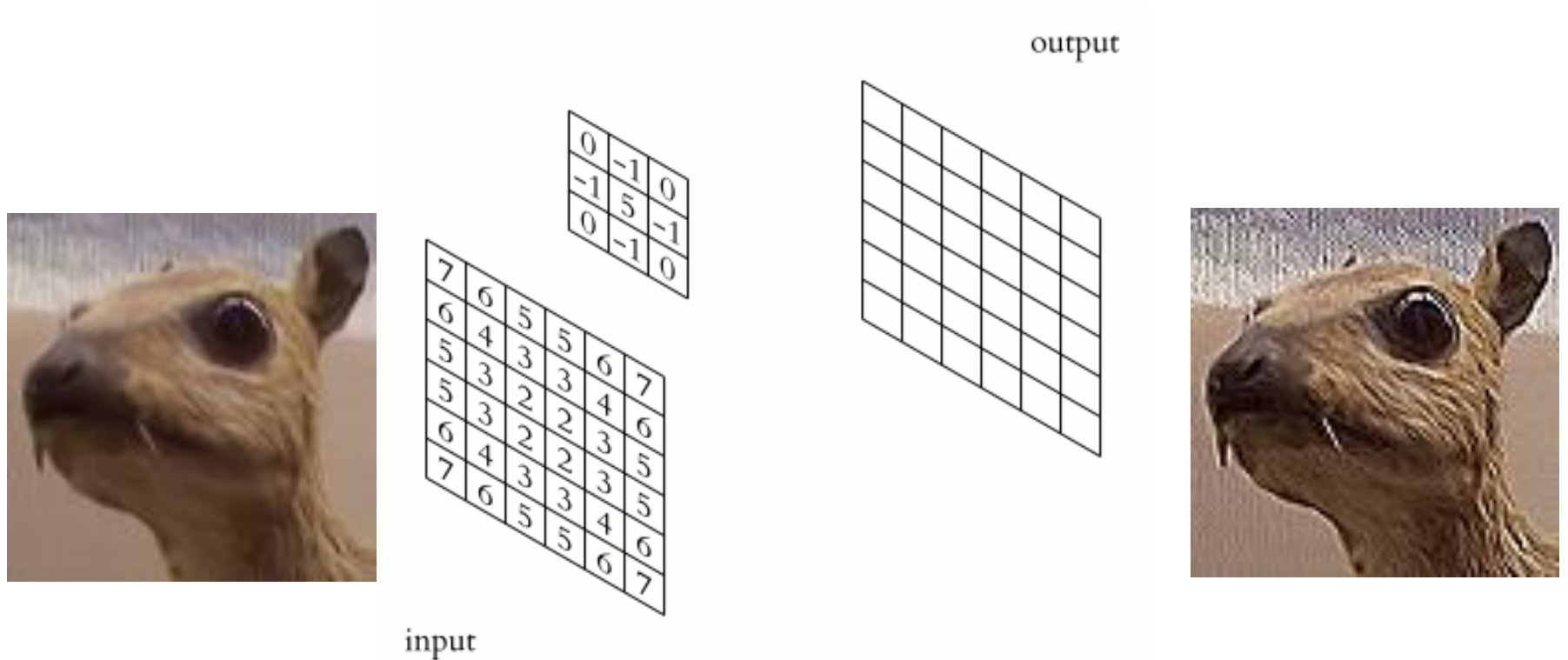


Image convolution



Sharpening filter

Corresponding point computation

- **Convolution** is crucial to build image filters for the estimation of corresponding points

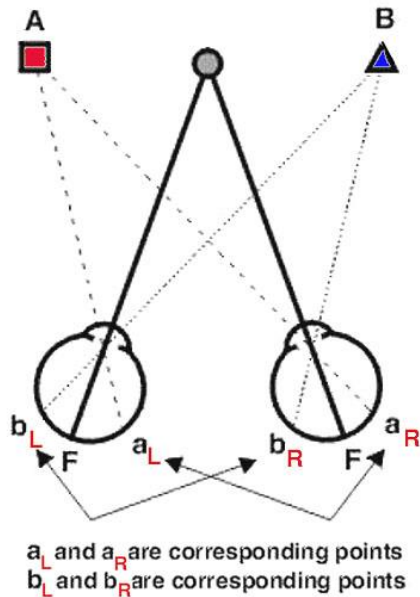
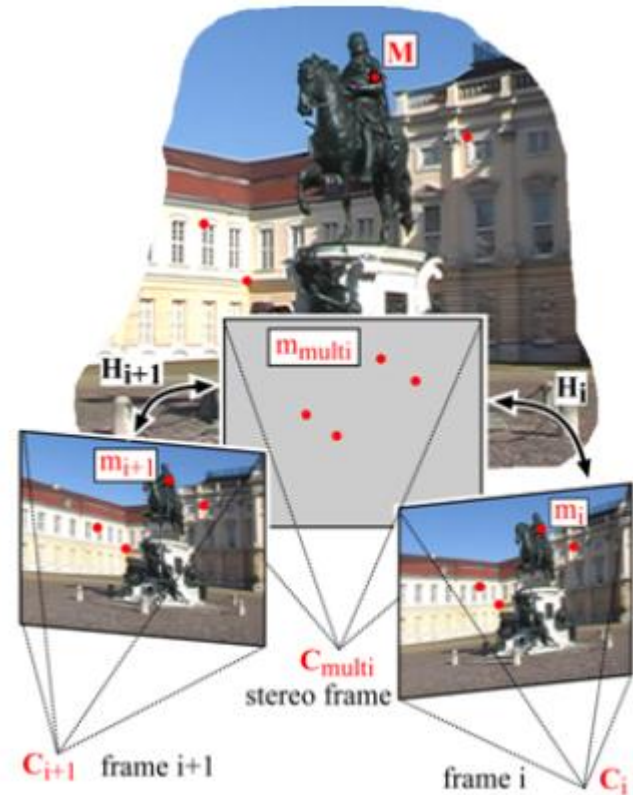
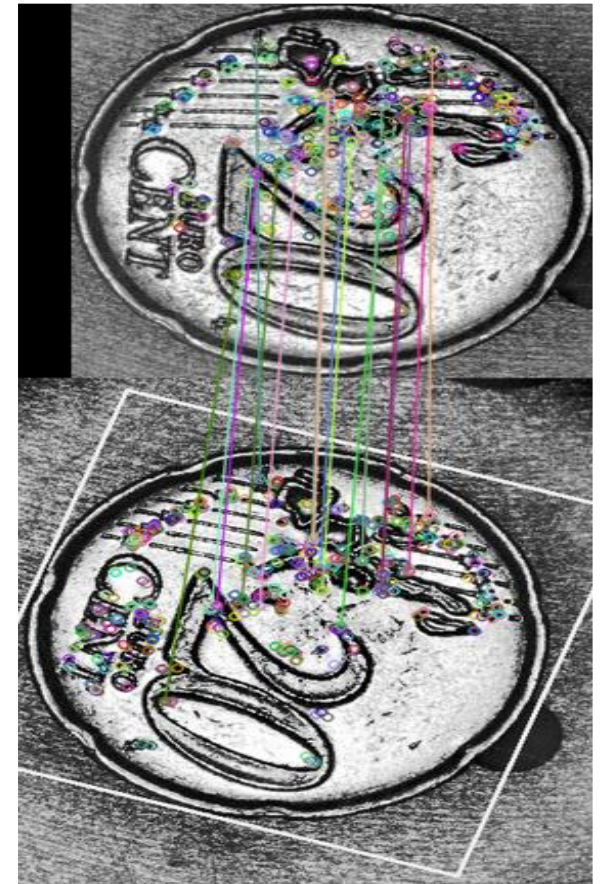


Figure 3. Corresponding points of the two eyes.



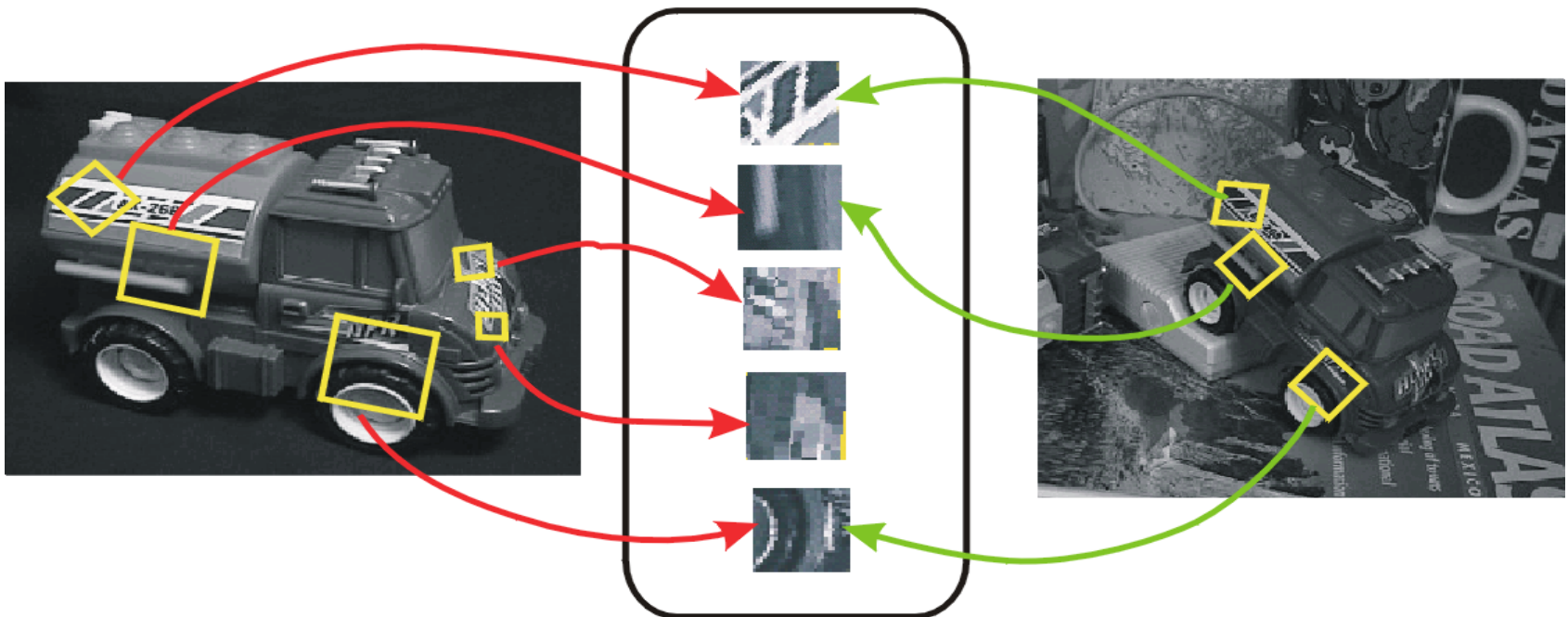
Corresponding point computation

- The idea consists to identify a limited set of **salient points** (or feature point) to be used for matching purposes.
- Three phases:
 1. Salient point detection,
 2. Salient point description,
 3. Salient point matching.



Scale Invariant Feature Transform

- The method Scale Invariant Feature Transform (SIFT) detects **salient points** (i.e., points with high contrast) using a **Laplacian of Gaussian (LoG)** filter to the **Scale Space** :



Scale Invariant Feature Transform

- The method must be invariant to:
 - **Scale,**
 - Rotation,
 - Illumination,
 - Point of view (affine).



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Scale Invariant Feature Transform

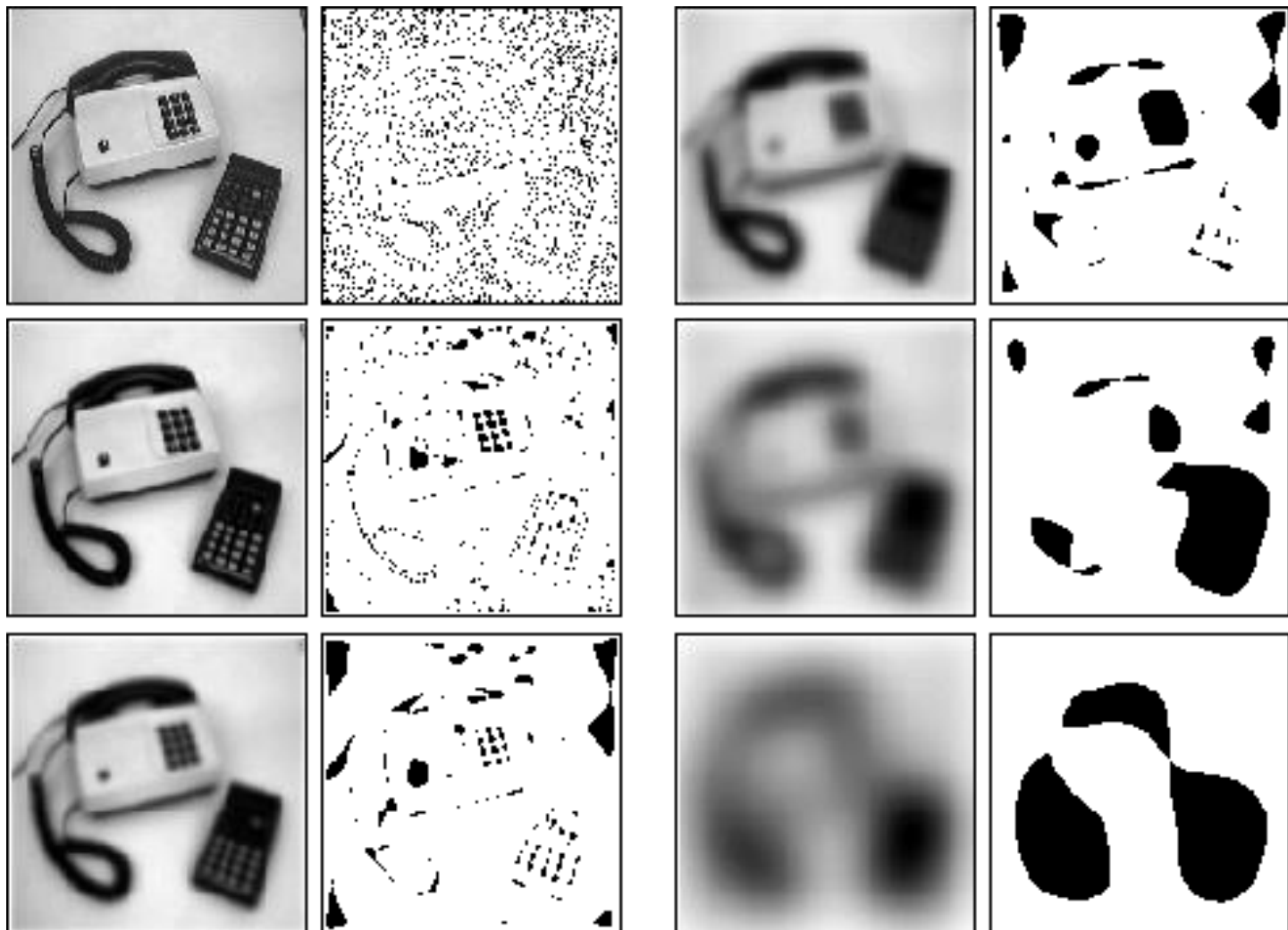
Objective: identify **location** and **scale** of *stable* points or regions. Two ideas: **Scale space** and **Image Pyramid**



The **Scale Space** of an image is a function $L(x, y, \sigma)$ that is obtained by a **convolution** of a Gaussian kernel (at different scales) with the input image:

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

Scale Space

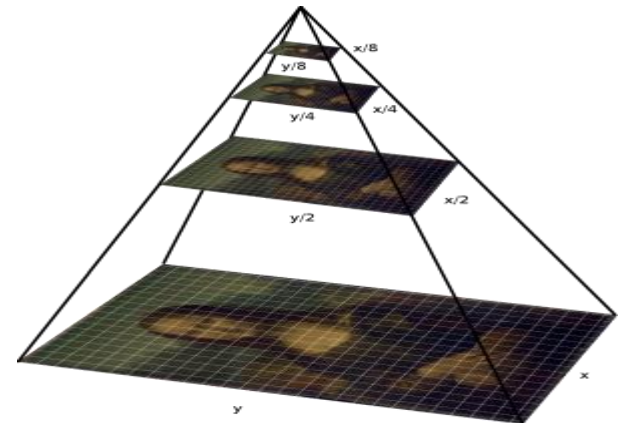


Scale Invariant Feature Transform

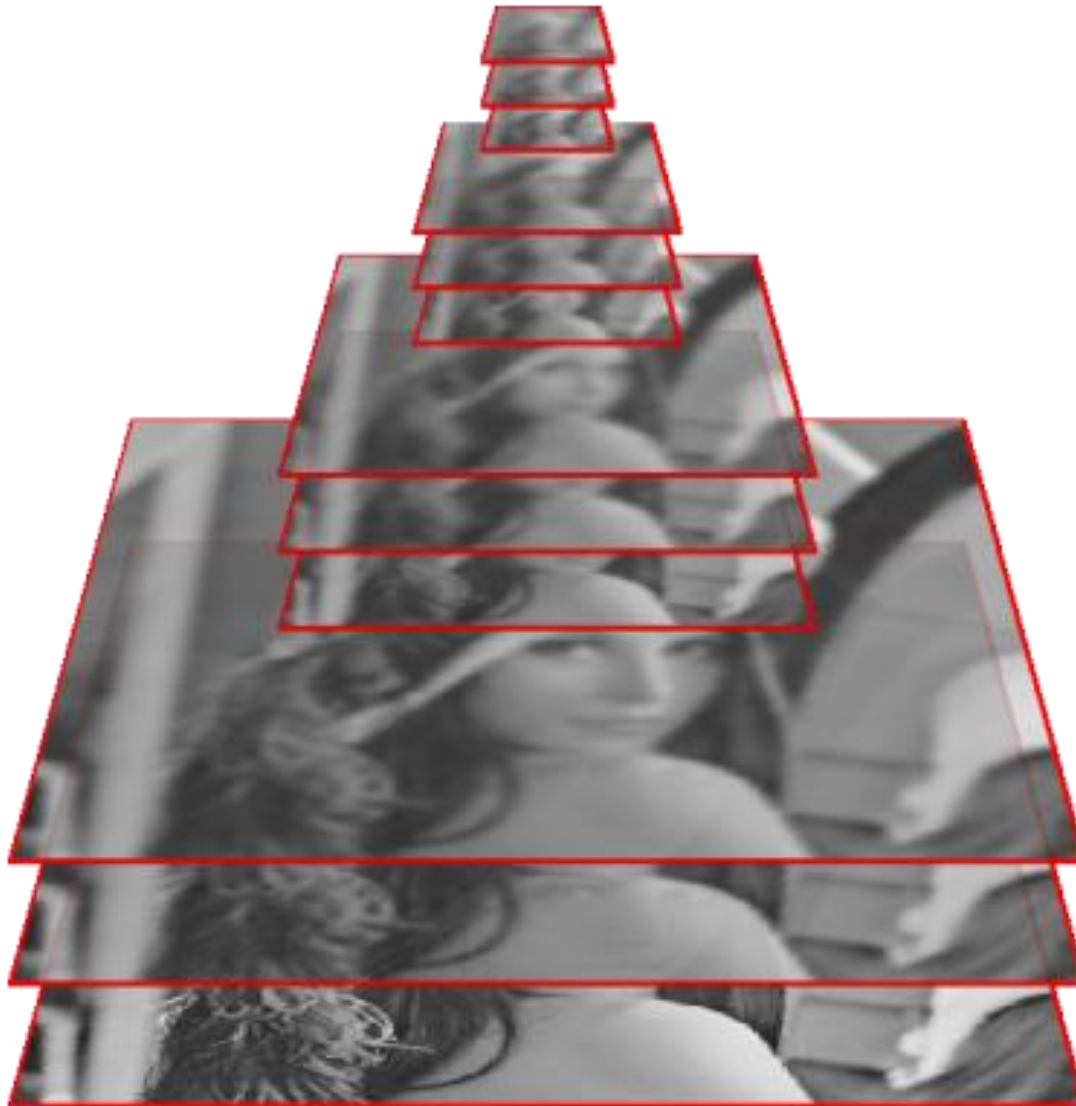
Objective: identify **location** and **scale** of *stable* points or regions. Two ideas: **Scale space** e **Image Pyramid**



Image Pyramid: it is a set of resampled images at different descendent levels (octave) :



Scale Space+ Pyramids



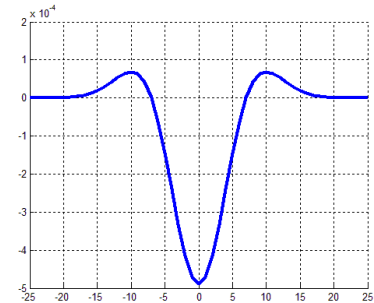
Scale Space+ Pyramids



Scale Invariant Feature Transform

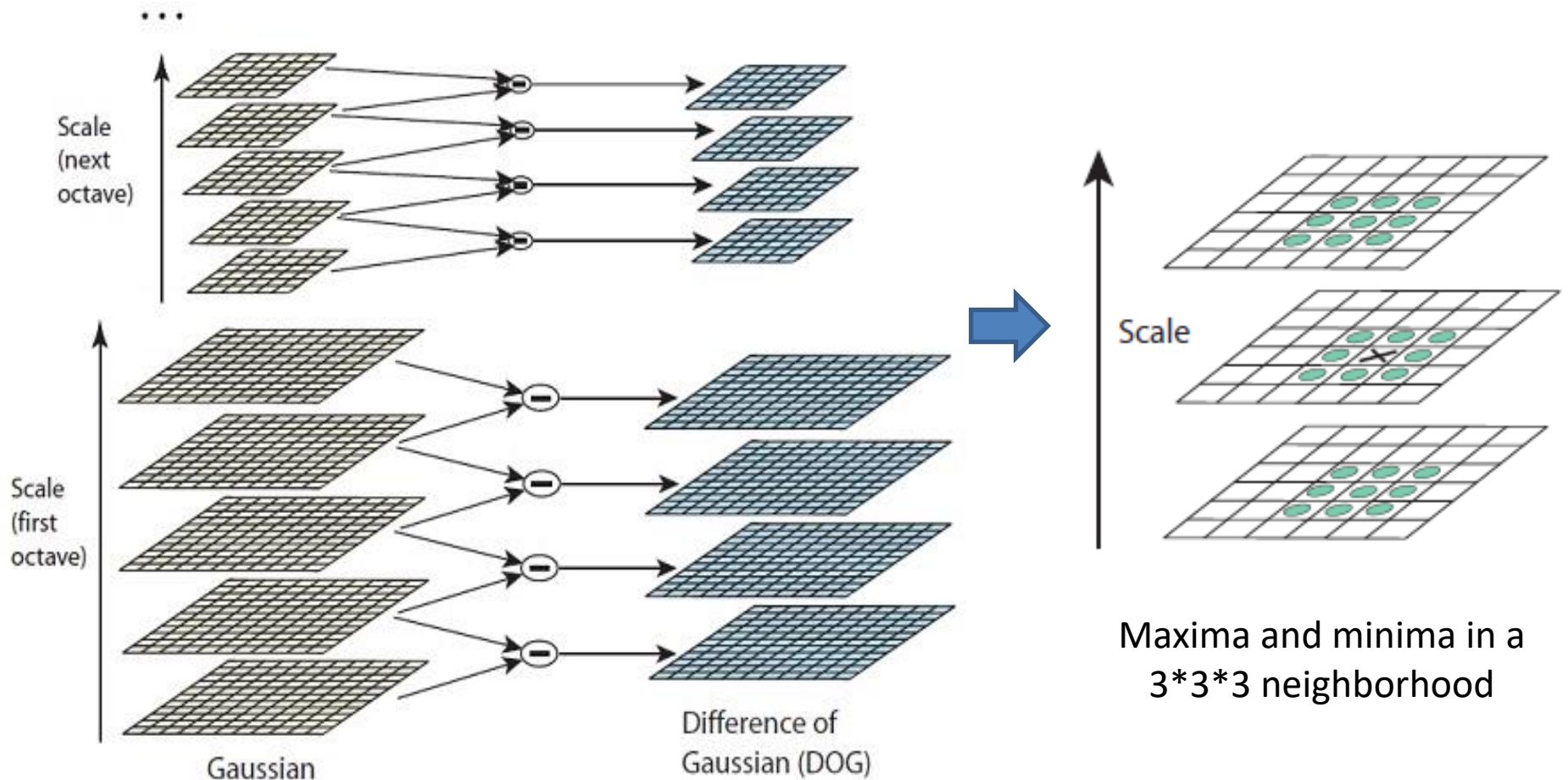
The **Difference of Gaussian** filter is used to approximate the **Laplacian of Gaussian** filter:

$$\begin{aligned} D(x, y, \sigma) &= (G(x, y, k\sigma) - G(x, y, \sigma)) * I(x, y) \\ &= L(x, y, k\sigma) - L(x, y, \sigma) \end{aligned}$$



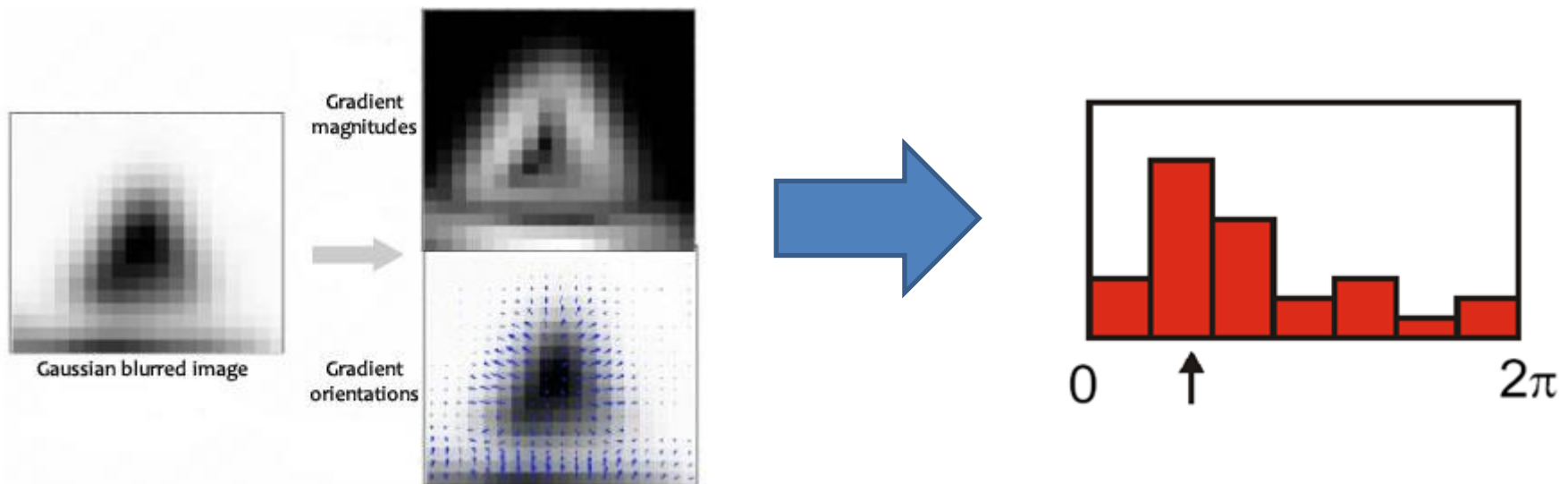
Salient points (blob) are extremants (min or max) for both space and scale (a 3x3x3 window is used on the scale space) !

Scale Invariant Feature Transform

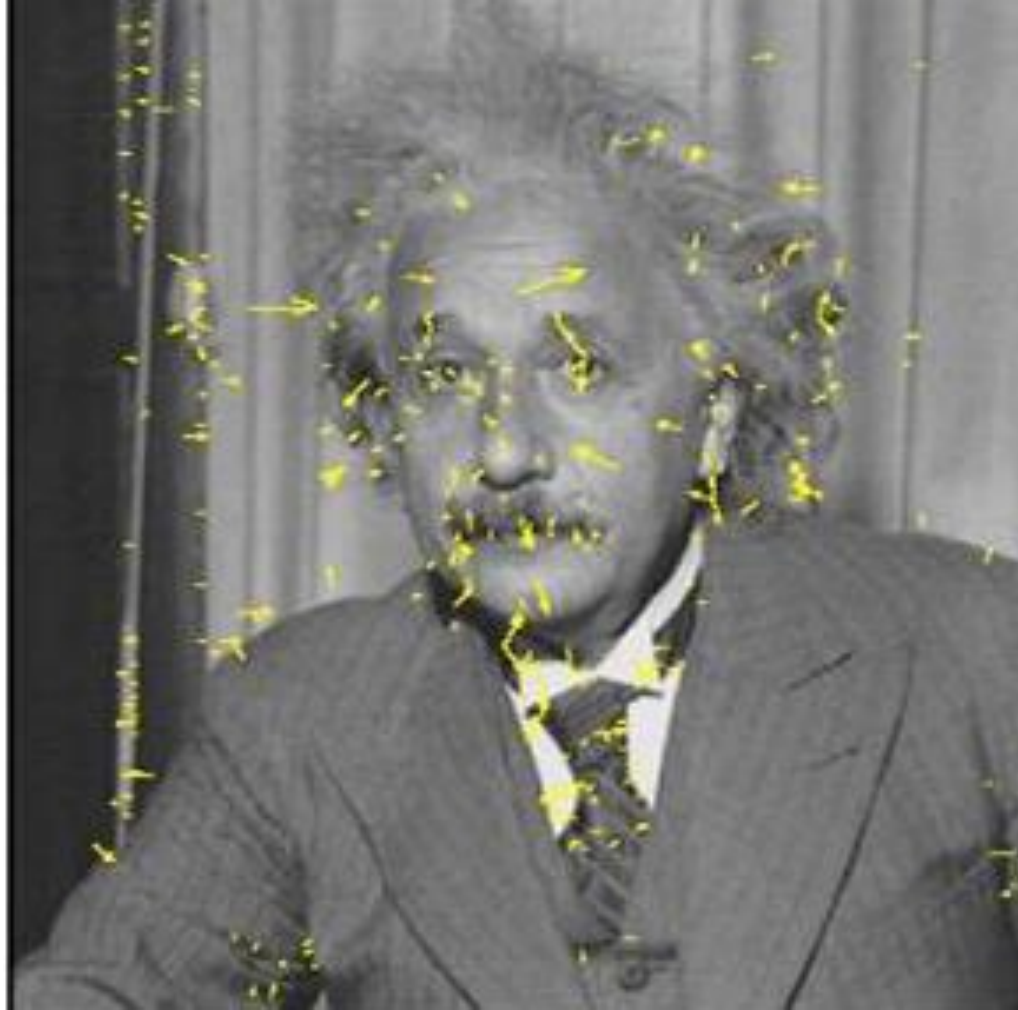


Scale Invariant Feature Transform

- **Orientaton:** it is estimated from the histogram of gradients on a local region of 16x16 pixels on the Gaussian pyramid level (scale) where the salient point has been detected.



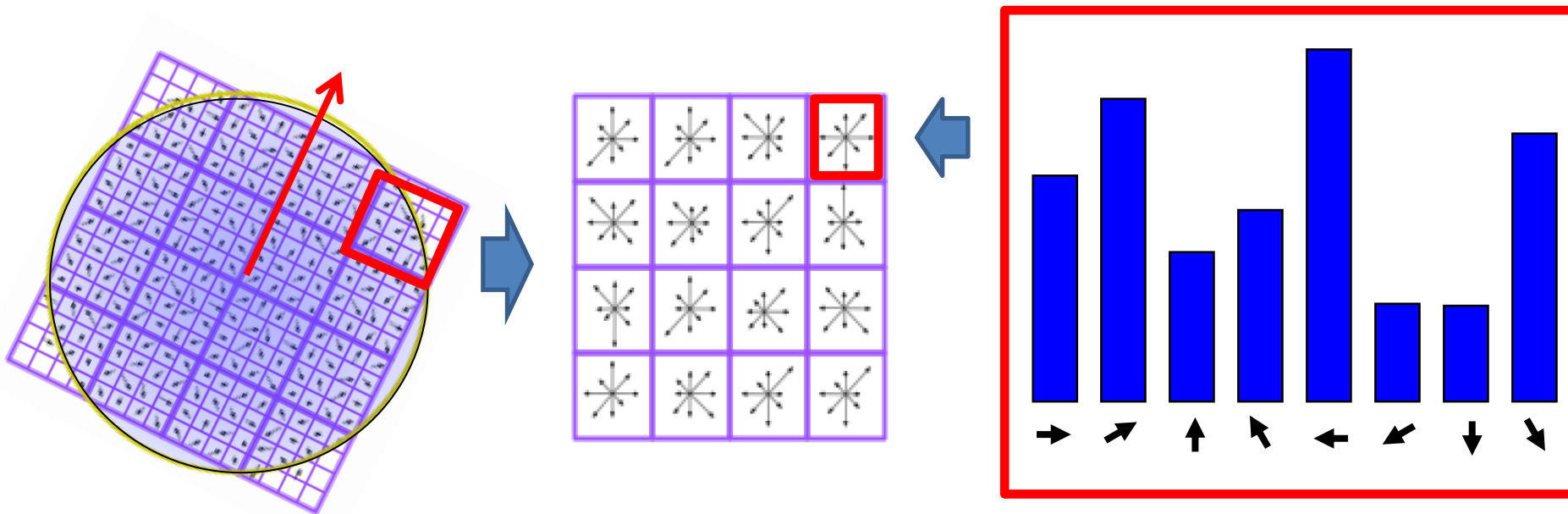
Scale Invariant Feature Transform



SIFT point detection: arrows represent orientation and length.

Scale Invariant Feature Transform

- **Point Descriptor:**



The point descriptor is a 128-vector obtained from the concatenation of the 16 histograms (8 bin each)!

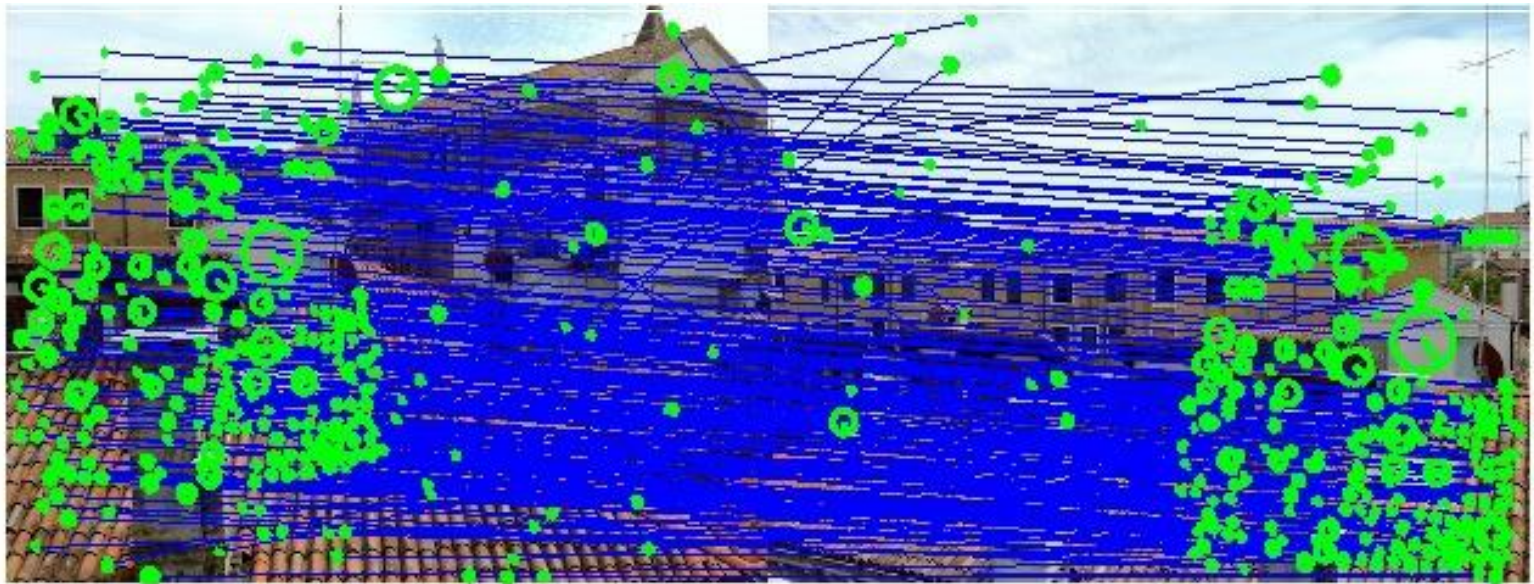
Scale Invariant Feature Transform

- The point matching is obtained from the comparison of descriptors

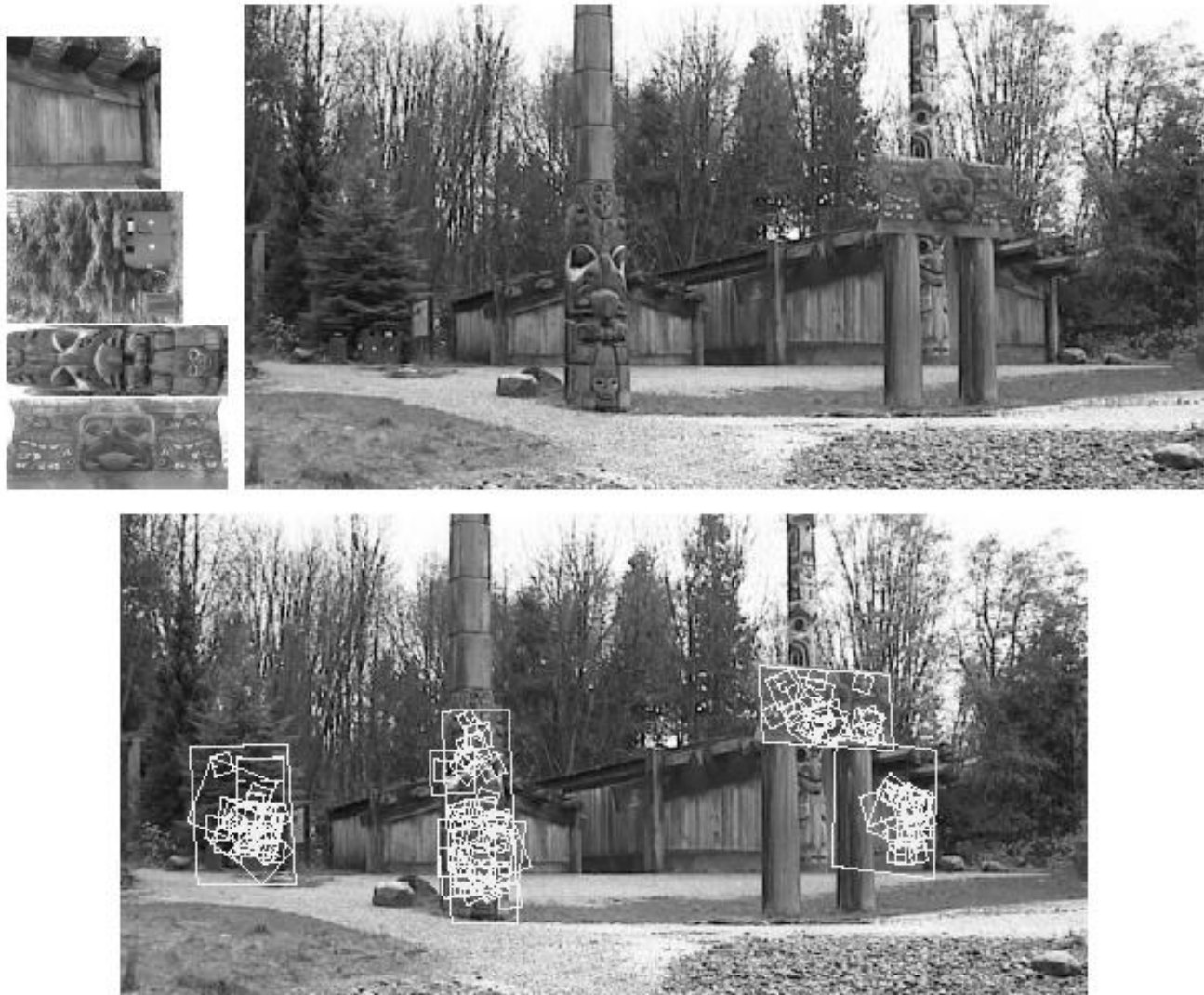


Scale Invariant Feature Transform

- The point matching is obtained from the comparison of descriptors

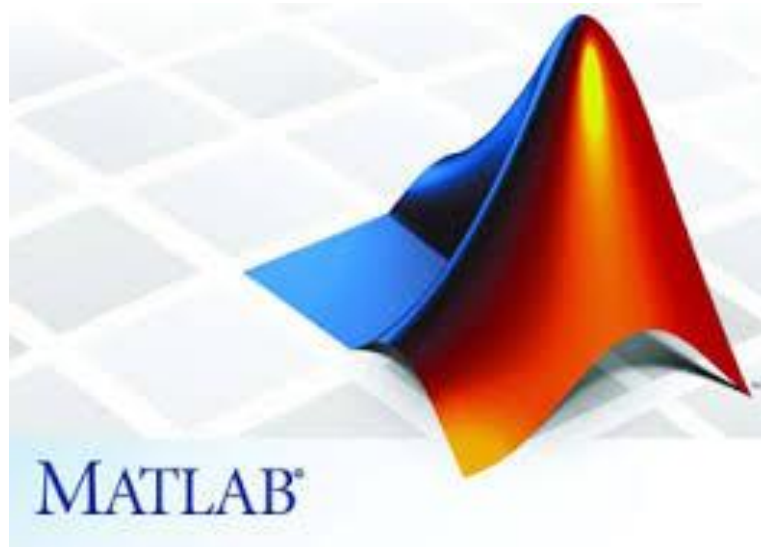


Scale Invariant Feature Transform



Scale Invariant Feature Transform

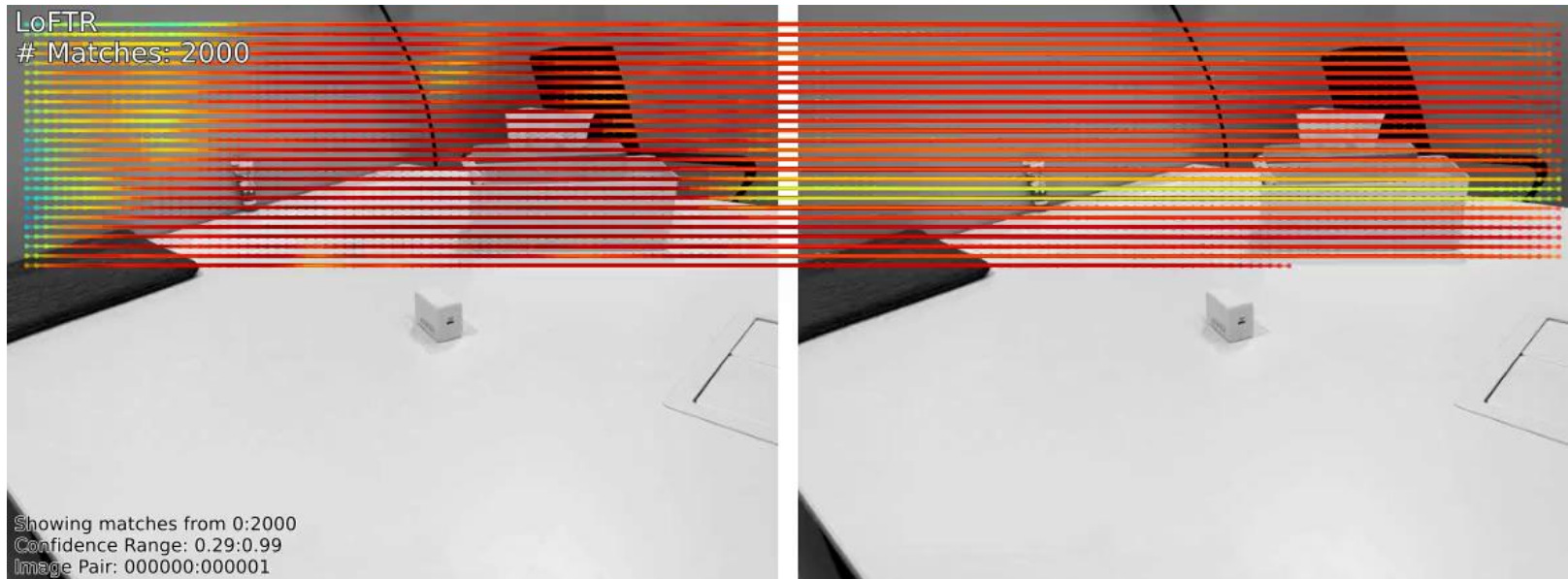
- Matlab Intermezzo:



<http://www.vlfeat.org/overview/sift.html#tut.sift.match>

New methods

- Nowadays feature matching can be obtained using deep learning methods,
 - E.g.: «LoFTR: Detector-Free Local Feature Matching with Transformers»



<https://zju3dv.github.io/loftr/>